Unit III

3. Internet Basics and Networking Protocol

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3.3 Networking protocol: IP, TCP, FTP, HTTP, DHCP

3.1 Internet- Architecture, Internet Service Providers (ISP), Internet Addressing System: IP Address, DNS, URL:

1. Internet Architecture:

The Internet is a **global network of interconnected computer networks** based on the **TCP/IP protocol suite**.

It consists of:

Clients: Devices like computers, mobiles that request services.
Servers: Provide services like websites, email, files.
Routers & Switches: Direct data traffic between networks.
Backbone: High-speed data lines that connect major networks.
Protocols: Govern communication (e.g., HTTP, TCP/IP, FTP).

2. Internet Service Providers (ISPs):

ISPs are companies that **provide access to the Internet** to individuals and organizations.
Examples: **Jio, Airtel, BSNL, ACT, etc.**
They offer services like broadband, mobile data, domain registration, web hosting, and email.

3. Internet Addressing System:

a) IP Address (Internet Protocol Address):

A unique **numerical identifier** for each device on a network.
Two types:

IPv4 (e.g., 192.168.1.1) – 32-bit address.
IPv6 (e.g., 2001:0db8:85a3::8a2e:0370:7334) – 128-bit address.
Used for **identifying and locating devices**.

b) DNS (Domain Name System):

Converts **human-readable domain names** (like google.com) into IP addresses.
Acts like a **phonebook of the Internet**.
Example: `google.com` → `142.250.64.78`

c) URL (Uniform Resource Locator):

The **web address** used to access resources on the Internet.
Format: `https://www.example.com/page.html`

`https` = protocol
`www.example.com` = domain name
`/page.html` = path to resource

In Summary:

The Internet relies on a layered architecture, ISPs for access, and an addressing system using **IP, DNS, and**

URLs to connect users to services and resources across the world.

3.2 Concept of Intranet & Extranet:

1. Intranet:

An **Intranet** is a **private network** used **within an organization** to share information, resources, and services among employees.

Features of Intranet:

Accessible **only to authorized users** within the organization.
Uses **Internet technologies** (like web browsers, servers, emails).
Improves **internal communication** and **collaboration**.

Examples:

Company HR portals, internal employee dashboards, internal document sharing systems.

Advantages:

Enhances productivity and communication.
Secure internal information sharing.
Centralized access to organizational resources.

2. Extranet:

An **Extranet** is an extension of the intranet that provides **controlled access to external users** such as **partners, suppliers, or customers**.

Features of Extranet:

Combines **intranet and internet technologies**.
Access is granted through **secure login systems** (VPN, firewalls, etc.).
Enables **B2B (business-to-business)** communication.

Examples:

Vendor portals, partner collaboration platforms, customer order tracking systems.

Advantages:

Strengthens external partnerships.
Enhances supply chain management.
Secure sharing of specific information with outsiders.

In Summary:

Intranet is used for **internal communication** within an organization.
Extranet allows **secure external access** to selected parts of the internal network, improving collaboration with partners and clients.

3.3 Networking protocol: IP, TCP, FTP, HTTP, DHCP:

1. IP (Internet Protocol):

IP is responsible for **addressing and routing** data packets between devices on a network.
Every device is assigned a **unique IP address**.

It ensures data is sent to the correct destination.
Types: **IPv4** and **IPv6**.

2. TCP (Transmission Control Protocol):

TCP ensures **reliable and error-free communication** between devices.
It breaks data into packets, sends them, and ensures they are **delivered in order**.
Commonly used with IP (TCP/IP).

3. FTP (File Transfer Protocol):

FTP is used to **transfer files** between computers over a network.
Allows uploading and downloading files from servers.
Requires **username and password** for access.
Example: Uploading website files to a hosting server.

4. HTTP (HyperText Transfer Protocol):

HTTP is used for **transmitting web pages** on the Internet.
It enables **communication between web browsers and web servers**.
Example: Accessing a website like `http://www.example.com`.

5. DHCP (Dynamic Host Configuration Protocol):

DHCP **automatically assigns IP addresses** to devices on a network.
Eliminates the need for manual IP configuration.
Ensures that no two devices have the **same IP address** (avoids conflict).

In Summary:

These protocols handle key networking tasks such as **data transmission (TCP/IP)**, **file sharing (FTP)**, **web access (HTTP)**, and **automatic IP assignment (DHCP)**, forming the backbone of communication on the Internet and local networks.